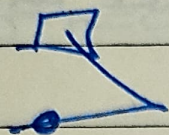


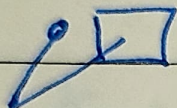
HIZP

Knee



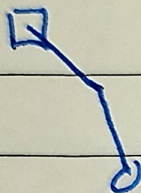
0

0



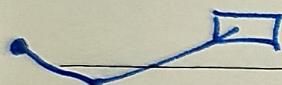
180

0



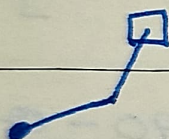
0

180



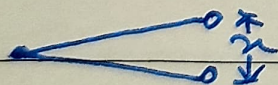
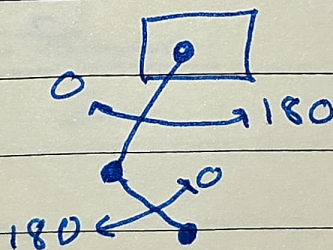
180

180

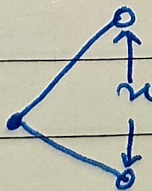


90

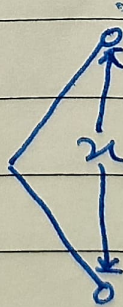
90



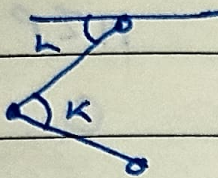
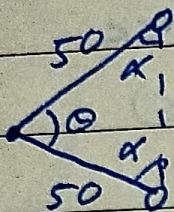
5/10



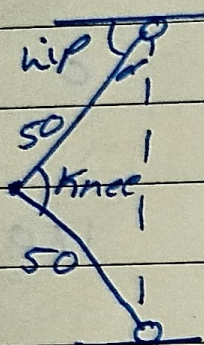
45/90



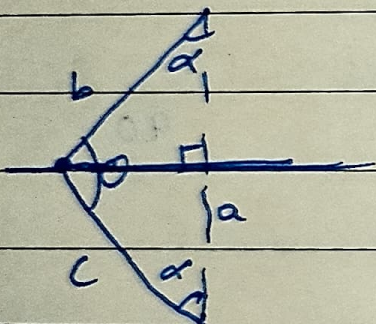
90/180



$h = \text{hip angle}$
 $k = \text{knee angle}$



height



$$\theta = \cos^{-1} \left[\frac{(b^2 + c^2 - a^2)}{2bc} \right]$$

$$A = \frac{180 - \theta}{2} = 90 - \frac{\theta}{2}$$

$$k_{nee} = \cos^{-1} \left[\frac{(50^2 + 50^2 - \text{height}^2)}{2 \times 50 \times 50} \right]$$

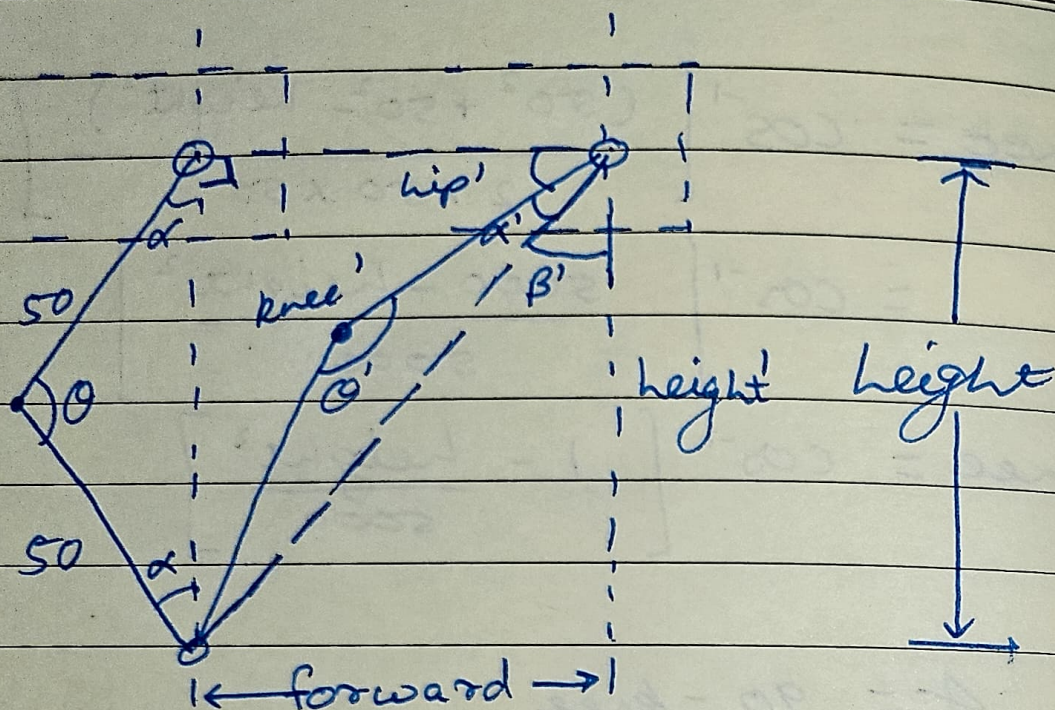
$$= \cos^{-1} \left[\frac{5000 - \text{height}^2}{5000} \right]$$

$$\therefore k_{nee} = \cos^{-1} \left[1 - \frac{\text{height}^2}{5000} \right]$$

$$A = \frac{90 - k_{nee}}{2}$$

$$\text{hip} = 90 - A = 90 - \left(\frac{90 - k_{nee}}{2} \right)$$

$$\therefore \text{hip} = \frac{k_{nee}}{2}$$



$$\text{height}' = \sqrt{\text{height}^2 + \text{forward}^2}$$

$$\text{knee}' = \theta' = \cos^{-1} \left(1 - \frac{(\text{height}')^2}{5000} \right)$$

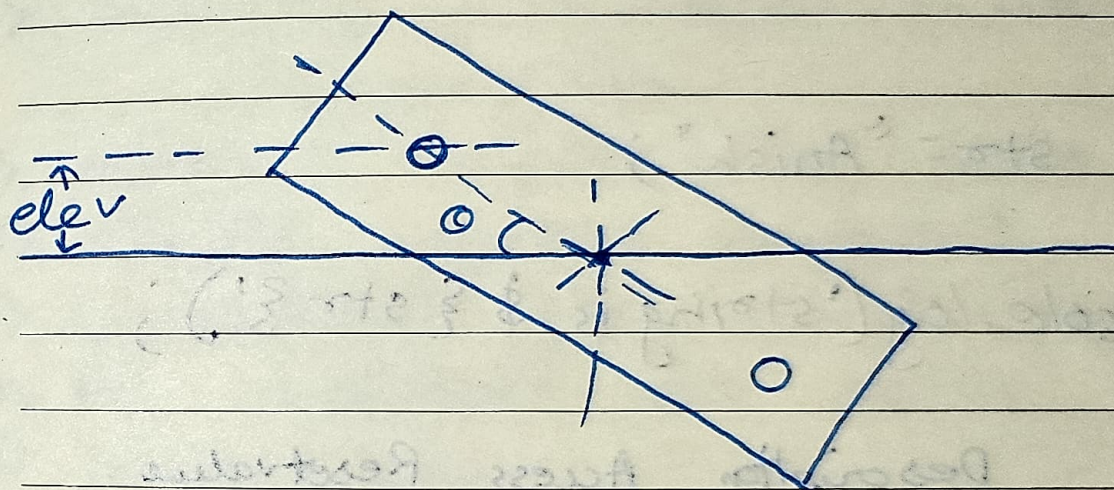
$$\text{hip}' = 90 - \alpha' - \beta'$$

$$\therefore \beta' = \tan^{-1} \left(\frac{\text{forward}}{\text{height}} \right)$$

$$\therefore \alpha' = \frac{180 - \theta'}{2} = 90 - \frac{\theta'}{2}$$

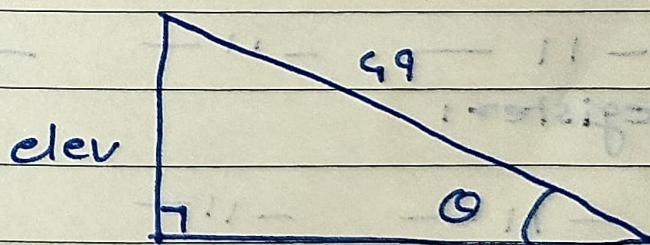
$$\text{hip}' = 90 - \left(90 - \frac{\theta'}{2} \right) - \tan^{-1} \left(\frac{\text{forward}}{\text{height}} \right)$$

$$\therefore \text{hip}' = \frac{\text{knee}'}{2} - \tan^{-1} \left(\frac{\text{forward}}{\text{height}} \right)$$



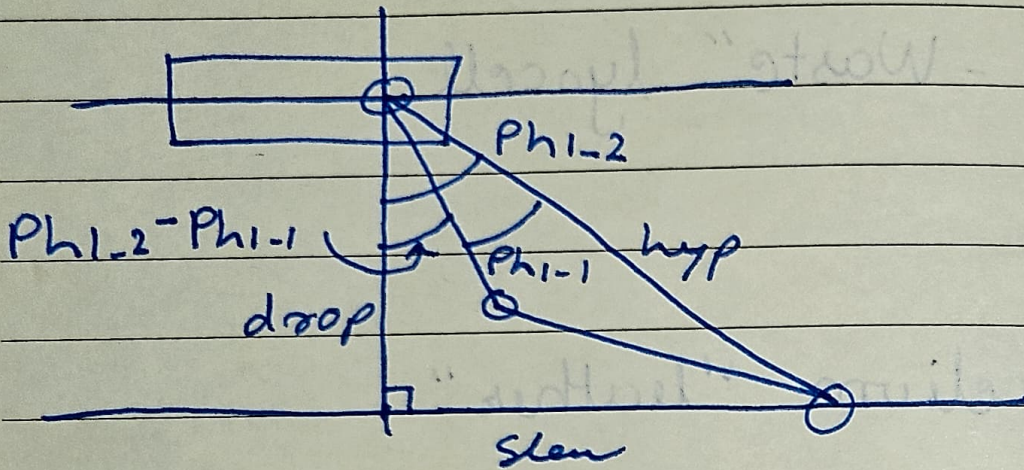
$\pm \text{elev}$

$\pm \theta$



$$\sin \theta = \frac{\text{elev}}{49} \quad \therefore \theta = \sin^{-1} \left(\frac{\text{elev}}{49} \right)$$

Inverse kinematics

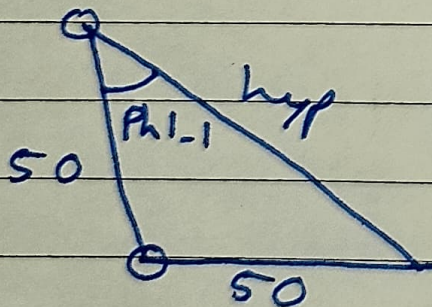
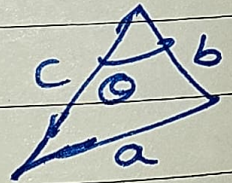


$$drop = height - required_height$$

$$hyp = \sqrt{s_{len}^2 + drop^2}$$

by cosine rule

$$\theta = \cos^{-1} \left[\frac{b^2 + c^2 - a^2}{2bc} \right]$$

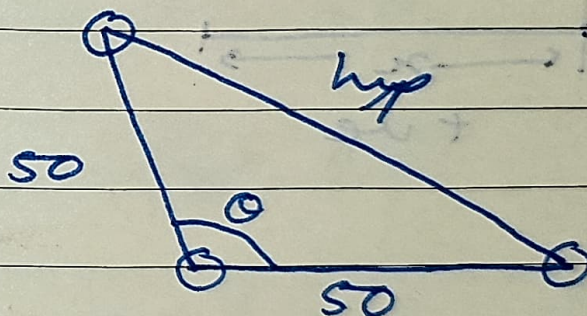


$$\phi_{1-1} = \cos^{-1} \left[\frac{\text{hyp}^2 + 50^2 - 50^2}{2 \times \text{hyp} \times 50} \right]$$

$$\therefore \phi_{1-1} = \cos^{-1} \left[\frac{\text{hyp}^2}{100 \text{ hyp}} \right]$$

$$\therefore \phi_{1-2} = \tan^{-1} \left[\frac{\text{slew}}{\text{drop}} \right]$$

$$\text{hyp} = 90 + (\phi_{1-2} - \phi_{1-1})$$



$$\text{knee} = \cos^{-1} \left[\frac{50^2 + 50^2 - \text{hyp}^2}{2(50)(50)} \right]$$

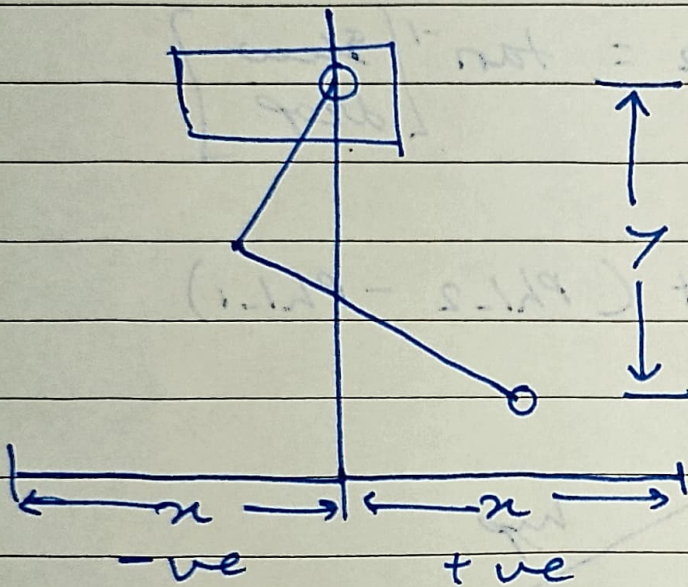
$$\therefore \text{knee} = \cos^{-1} \left[1 - \frac{\text{hyp}^2}{5000} \right]$$

one foot

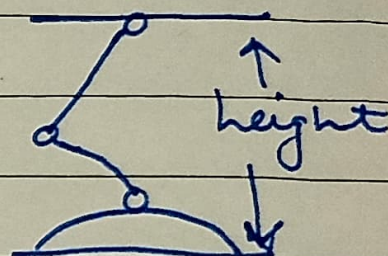
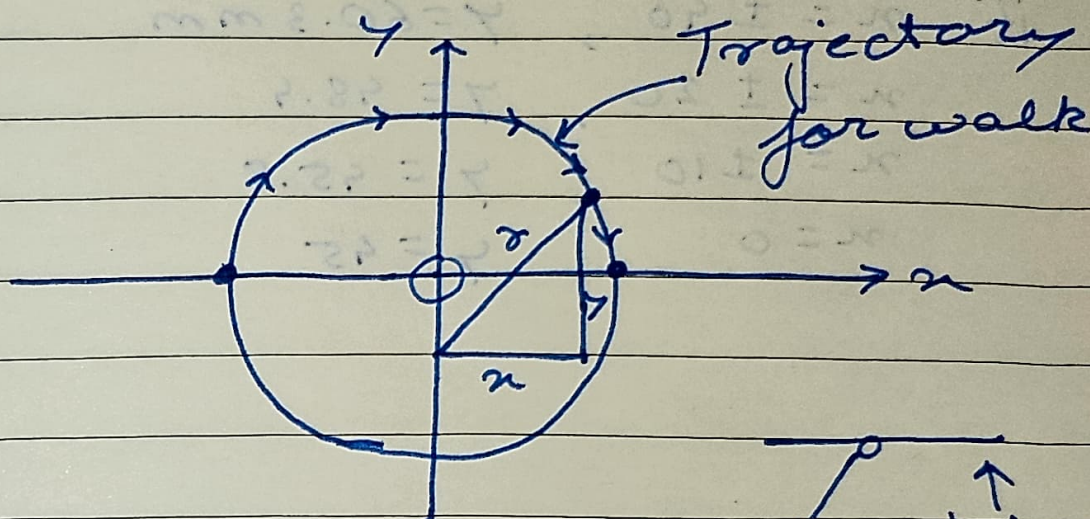
3 • ↑ 1
2 ↑ • 4

two foot

↑ •
• ↑



For walk



$$r^2 = x^2 + y^2$$

$$y^2 = r^2 - x^2$$

$$y = \sqrt{r^2 - x^2}$$

drop
offset
-150
mm

$$y = \left[\sqrt{r^2 - x^2} \right] - 105$$

$$r = 60 \text{ mm}$$

$$\text{if } n = \pm 40, \quad \gamma = 60.3 \text{ mm}$$

$$n = \pm 20, \quad \gamma = 48.4$$

$$n = \pm 10, \quad \gamma = 45.8$$

$$n = 0, \quad \gamma = 45$$

Front

left

Right

Back